



THE BOXES

Research dossier

A drama set inside NASA Mission Control across the four days of Apollo 13, from the oxygen-tank rupture to splashdown in April 1970.

The research reconstructs the interior operations of NASA Mission Operations Control Room 2 in Building 30 across four days in April 1970. Ground teams manage telemetry dropouts, life support limits, and trajectory burns after an oxygen tank ruptures on Apollo 13. The story stays inside the room with the engineers and controllers who missed earlier test discrepancies. These men must resolve the emergency while confronting their own procedural errors.

Depth	Production
Research confidence	77%
Coverage	85%
Status	—
Boxes	13
Evidence fragments	241
Distinct sources	186
Distinct domains	72
Contradictions	0 conflict, 0 context
Research passes	4
Generated	2026-09-09 15:36 UTC
Project id	demo

THE BOXES

What the research found, in plain language, box by box. Every claim below is traceable to the sources listed under it.

PRESS CORPS INFRASTRUCTURE

99%  8 items · 5 domains

Press operations occupy the auditorium within Building 1 adjacent to the control facility. Television cameras sit on elevated wooden risers facing the main presentation stage. Rows of teletype machines and banks of coin-operated payphones line the media workspace. Briefing officers distribute paper mission change sheets to reporters after major shifts.

- Building on a Mission: The Auditorium and External Relations Office Building - NASA · 2023-09-28
- News Media Handbook · 2005-02-01
- The display/control complex of the Manned Space Mission Control ...
- The Press Site at KSC · 2006-11-29
- Mission Control Center (MCC) - Apollo 13 - Fourth (4th ...
- APOLLO 13 MISSION REPORT SEPTEMBER 1970

OFFICE OF CIVILIAN DEFENSE RELATIONS

98%  12 items · 6 domains

Federal security details coordinate with local police departments to protect the space center perimeter. Armed guards monitor gate traffic and control access points around sensitive technical buildings. Administrative offices maintain contact records with civilian law enforcement during the Vietnam War period. Security officers log perimeter disturbances and visitor movement around federal assets.

- Records - Houston Police Department
- [PDF] Police Practices in Houston, Texas - U.S. Commission on Civil Rights
- Records transmittal and receipt from NASA Manned Spacecraft Center to Federal Records Center, GSA, Fort Worth, Texas for Record Group 255
- Historical Documents - Office of the Historian · 2026-06-29
- Building on a Mission: The Houston Mission Control Center - NASA · 2023-09-23
- Research in Vietnam War Records
- Houston Police Department - Public Information Requests
- Vietnam War: The War at Home | National Archives · 2022-06-17
- News Release Archives
- Houston Police Department
- Transparency

REAL-TIME TELEMETRY DATA FLOW · emergent

97%  14 items · 7 domains

Spacecraft tracking data flows from overseas antennas into the Goddard Space Flight Center in Maryland. Mainframe computers process incoming telemetry signals and route them directly to Houston. High-speed lines drive the numeric displays on console screens and front-wall plotters. Technicians track real-time coordinates against fixed mission trajectory curves.

- MANNED SPACE FLIGHT NETWORK FOR APOLLO - MIT · 2002-10-31
- FLIGHT-CONTROL DATA NEEDS, TERMINAL DISPLAY ... · 2005-06-09
- Apollo Era Technical Description
- Deep Space Network - NASA Science · 2025-01-16
- Integrated telemetry-computer system
- Telemetry and Command (T&C) Systems · 2026-03-13
- https://tda.jpl.nasa.gov/progress_report/42-25/25C.PDF
- NATIONAL AERONAUTICS AND SPACE ADMINISTRATION 1971 15,
- 1970 December 27 ~(C
- [PDF] Real-Time Flight Test Data Distribution and Display
- Network Space Deep The 7, Progress'Report.42-40
- SECTION 7 CRT DISPLAY PAGES
- -

ENGINEERING DIVISION DEMOGRAPHICS

97%  15 items · 5 domains

The flight controllers on duty are largely young civilian engineers in their twenties and thirties. Many staff members hold engineering degrees from regional universities across the American South and Midwest. Several positions are filled by military veterans with systems maintenance backgrounds. The room operates through assigned stations including flight dynamics, guidance, and environmental control.

- Information Summaries Astronaut Fact Book - NASA · 2013-04-08
- Astrona · 2008-05-23
- Apollo 13 - NASA · 2023-03-29
- Apollo 13 Archives - NASA
- List of NASA's flight control positions - Wikipedia · 2026-06-21
- Flight controller - Wikipedia · 2026-06-29
- of
- Documentary Spotlights Apollo-Era NASA Flight Controllers – Houston Public Media · 2017-04-15
- Msc Personnel
- Apollo 13 II - NASA · 2023-07-06
- biographical summaries

CONSOLE HARDWARE ARCHITECTURE

96%  16 items · 9 domains

Building 30 contains two Mission Operations Control Rooms on the second and third floors. Consoles house cathode-ray tube displays alongside illuminated push-button indicators for systems monitoring. Pneumatic tube stations move paper telemetry summaries and schematics across the room. Overhead projection screens display spacecraft coordinates and trajectory plots above the console rows.

- Apollo Mission Control Center Restoration - NASA · 2026-04-08
- NASA's restored Apollo Mission Control is a slice of '60s life, frozen in amber - Ars Technica · 2025-05-16
- Apollo Mission Control Photos - Space Center Houston · 2022-09-23
- MOCR - Manned Spaceflight Operations Association · 2018-03-13
- Orbiter Displays Reference Guide (OI-34)
- AS-508 MISOO FIU A IN
- MCC Operational Configuration Mission J1 AS-510/SC112/LM1C Apollo 15
- COMMAND AND SERVICE MODULE CONTROLS AND ...
- mission operations center: Topics by Science.gov
- CRT Reference Summary OI-27
- Microsoft Word - Final Drft, MCC 2010-12-30
- [PDF] Apollo Mission Control Restoration - NASA
- GitHub - johnmknight/MOCR-Console-Replica-Suite: Period ...
- The display/control complex of the Manned Space Mission Control ...

COMMUNICATIONS LOOP ACOUSTICS

96%  17 items · 11 domains

Controllers monitor separate voice channels covering the flight director loop, spacecraft communications, and technical support backrooms. Tapes preserve clipped verbal cadences, overlapping readouts, and background hum from tracking ground stations. Transmission fidelity changes when signals switch between worldwide receiver sites. Transcripts document immediate verbal responses to warning tones and telemetry drops.

- Apollo 13, transcription du vol
- Mission Control Audio Channel
- Apollo 13 in Real Time
- Ben Feist Interview on Apollo 13 Audio | Everyday Astronaut · 2020-11-23
- Apollo 13 Transcripts
- How Mission Control saved Apollo 13 | BBC Sky at Night Magazine · 2026-06-15
- Mission Transcripts: Mercury, Gemini, and Apollo - NASA · 2026-06-15
- NASA Audio Highlight Reels, Soundbites and Launch Sounds
- Apollo 13 Audio

- NASA, university researchers release 19000 hours of ... · 2018-07-25
- Apollo 11 : NASA : Free Download, Borrow, and Streaming · 2010-12-07
- Recovering Audio from the Apollo Moon Missions
- Apollo 11 - Technical Air-to-Ground Voice Transcription

CLEAR LAKE COMMUTING

95%  12 items · 7 domains

Controllers commute to the Clear Lake complex along local Texas access highways and regional roads. Personnel park personal sedans and pickups in open asphalt lots outside the security perimeter. Security personnel check identification credentials at gated access booths. The facility sits isolated twenty-five miles southeast of downtown Houston.

- 309 Clear lake Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- 19 Clear lake area, Nasa Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- Clear Lake - Police - City of Houston
- 5 Lct area, 1970 s Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- ROAD SURFACES AROUND NASA, NASA Technology Images — PICRYL - Public Domain Media Search Engine
- 60 Years Ago: NASA Selects Houston as Site for New Manned Spacecraft Center - NASA · 2023-08-30
- Microsoft Word - If99-6723_final-AVT-152-005_NASA_Experience_w-Pogo_in_Human_Spaceflight_Vehicles_Final.doc
- 104 Clear lake, Texas Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- 100 Clear lake, Houston Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- City of Houston eGovernment Center > About Houston > NASA
- CLEAR LAKE No. 81
- MAP OF THE CLEAR LAKE AREA

TECHNICAL CRISIS VOCABULARY

94%  10 items · 6 domains

Flight teams speak in abbreviated operational acronyms to manage telemetry data quickly. Controllers refer to flight positions as Trench, GUIDO, and CAPCOM during anomalies. Numerical values for power budgets and consumable percentages are called out in standard short-form code. Telemetry failure reports follow rigid verbal checklists established in malfunction procedure manuals.

- [PDF] - 3 GLOSSARY OF ACRONYMS AND ABBREVIATIONS - Ibiblio
- GLOSSARY OF TERMS · 2010-01-27
- Appendix B: Acronyms, Abbreviations, and Symbols - NASA · 2026-04-22
- Apollo 13 LM Malfunction Procedures
- [PDF] _.-_m MANNED SPACECRAFT CENTER - NASA
- NASA-Apollo-Mission-Control-Acronyms. ...
- Space Transportation System and Associated Payloads: Glossary, Acronyms, and Abbreviations
- Space Flight Technical Language Handbook
- APOLLO 13 MISSION REPORT SEPTEMBER 1970

FACILITY LIGHTING PROTOCOLS

94%  10 items · 9 domains

The Mission Operations Control Room maintains low ambient lighting levels to protect screen legibility. Overhead fluorescent banks are dimmed to prevent glare on glass cathode-ray tube monitors. Front projection screens require dark perimeter walls to keep trajectory graphics readable. Individual consoles use hooded task lights to illuminate paper logs without washing out screens.

- PPF to Foot-candle Reference Table
- Foot-candle - Wikipedia · 2026-03-08
- FOOTCANDLE LIGHT GUIDE - Lighting Design Lab · 2013-07-29
- Foot Candle Calculator - Lighting Requirements & Illuminance ...
- https://www.usmd.edu/usm/adminfinance/Facilities/forms/interior_lighting_fluorescent.pdf
- Lighting Illumination Calculator - everycalculators.com
- Apollo Flight Controller 101: Every console explained - Ars Technica · 2019-12-25
- 3.PDF

MANUAL CALCULATION TOOLS · emergent

89%  14 items · 5 domains

Engineers rely on manual calculation tools when mainframe telemetry channels drop offline. Controllers calculate burn times and orbital changes using Pickett pocket slide rules. Teams mark reference trends on printed grid paper using grease pencils. Desk calculators and printed trajectory tables provide manual verification of remaining electrical reserves.

- Apollo Command Module Block II Primary Guidance, Navigation, and Control System Manual · 2018-07-27
- Quasi Official Apollo 13 Paper September 8.ppt · 2009-09-08
- - · 2004-09-22
- Slide Rule, 5-inch, Pickett N600-ES, Apollo 13 | National Air and Space Museum
- Don't Know What a Slide Rule is For | National Air and Space Museum
- STUDY OF CONTROL COMPUTERS FOR ...
- AS-508 MISOO FIU A IN
- E-2448 Rev 1, Users' Guide to Apollo GN&CS Major Modes and Routines
- 1970 Slide Rule
- Slide rule vintage hi-res stock photography and images
- ORBITAL TESTING REQUIREMENTS FOR GUIDANCE AND CONTROL DEVICES

CONTROL ROOM CONSUMABLES

62%  38 items · 13 domains

Mission teams rely on vending machines, institutional cafeteria deliveries, and diner takeout brought into the facility. Cardboard coffee cups and metal vacuum pots sit directly beside console control panels. Ash receptacles stand beside technical logbooks on console writing ledges. Paper wrappers and fast-food cartons accumulate on desks during continuous shift rotations.

- 93 Space food, Nasa Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- 120 Food containers, Nasa Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- Food Services - NASA · 2026-09-04
- NASA Facts
- [PDF] LYNDON B. JOHNSON SPACE CENTER
- Apollo Mission Control Photos - Space Center Houston
- 130 Food container, Crew Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- 132 Food container, Activities Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- 223 Spacecrew supplies Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- Apollo Mission Control Restoration - NASA
- APOLLO EXPERIENCE REPORT
- Apollo 13 Flight Journal - Mission Documents
- N76 12690 · 2004-01-23
- 163 Food container, Nasa Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- 100 Space food, Food Images: PICRYL - Public Domain Media Search Engine Public Domain Search

SHIFT ROTATION ATTIRE

58%  27 items · 12 domains

Controllers remain at their consoles across consecutive operational shifts spanning up to eighty-four hours. Photographic records show short-sleeve white dress shirts wrinkling over multiple days. Neckties are loosened and collar buttons unfastened as shifts progress. Ashtrays on console surfaces fill with discarded cigarette butts during continuous duty cycles.

- Mission Control, Houston, April 13, 1970 - NASA · 2015-04-13
- Apollo 13: The Successful Failure · 2020-04-06
- Apollo 13 - Wikipedia · 2026-08-13
- Apollo 13
- File:1970 Mission Control Apollo 13.jpg - Wikimedia Commons
- How Mission Control saved Apollo 13 - BBC Sky at Night Magazine

- PHOTOGRAPHS WRITTEN HISTORICAL AND ...
- 165 Flight controllers, Astronauts Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- 992 Controllers Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- 252 Consoles, Control Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- Apollo 13, NASA Apollo program. NASA public domain image... — PICRYL - Public Domain Media Search Engine
- Project Apollo Archive's albums | Flickr
- skandalon
- 43 Flight clothing, Astronauts Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- I found an archive of every picture taken by the astronauts ...

SUPPORT STAFF TRENCH DYNAMICS · emergent

26%  32 items · 13 domains

Staff Support Rooms sit directly behind the front trench to provide analytical calculations. Specialist engineers review subsystem data books and relay recommendations up the command chain. Teams work at grouped tables surrounded by blueprinted schematics and paper printouts. The backroom groups calculate vehicle margins before flight controllers commit to burns.

- Apollo experience report - real-time auxiliary computing facility development · 2002-04-17
- Mission Control, Houston · 2006-05-01
- Going boldly: Behind the scenes at NASA's hallowed Mission Control Center - Ars Technica · 2012-11-14
- The Trench: The People at Mission Control | American Experience | Official Site | PBS · 2017-12-28
- Diagram of NASA Mission Control | National Air and Space Museum
- Apollo Mission Control Center Restoration · 2026-04-08
- The newly restored Apollo Mission Control Room : r/nasa
- Background Data (Process Overview)
- FLIGHT-CONTROL DATA NEEDS, TERMINAL DISPLAY ...
- MANNED SPACE FLIGHT NETWORK FOR APOLLO - MIT
- MCC Operational Configuration Mission J1 AS-510/SC112/LM1C Apollo 15 · 2002-02-05
- Mission Operations Directorate New Employee Orientation
- Apollo 13
- Apollo 13 (film)
- Apollo 13, We Have a Solution: Part 2 - IEEE Spectrum · 2021-07-25

CONTRADICTIONS

Where two sources disagree. Similarity alone never decides this; each pair below was read and judged.

None found.

REFERENCE REEL

The strongest material, cut into a sequence a director can read.

00:00 Establishing Mission Control

Archival materials document the control room environment during Apollo 13 in Houston.

- Mission Control, Houston, April 13, 1970 - NASA · nasa.gov
- Apollo Mission Control Center Restoration - NASA · nasa.gov · 2026-04-08
- Going boldly: Behind the scenes at NASA's hallowed Mission Control Center - Ars Technica · arstechnica.com · 2012-11-14
- How Mission Control saved Apollo 13 | BBC Sky at Night Magazine · skyatnightmagazine.com · 2026-06-15

00:30 Multi-Channel Communications

Restored multi-track audio channels record simultaneous conversations across flight controller consoles.

- Mission Control Audio Channel · apolloinrealtime.org
- Ben Feist Interview on Apollo 13 Audio | Everyday Astronaut · everydayastronaut.com · 2020-11-23
- Apollo 13 in Real Time · apolloinrealtime.org

01:00 Air-to-Ground Crisis Logs

Technical flight transcripts record critical communications following the oxygen tank failure.

- Apollo 13, transcription du vol · capcomespace.net
- Apollo 13 Transcripts · apollojournals.org
- Apollo 13 Audio · dn801309.us.archive.org

01:30 Extended Crisis Strain

Real-time records capture the extended four-day rescue operation through splashdown.

- Apollo 13 in Real Time · apolloinrealtime.org
- Apollo 13 - NASA · nasa.gov

DEPARTMENTS

The same boxes, grouped by which crew briefs off them, with what each one actually contains.

Script

COMMUNICATIONS LOOP ACOUSTICS · 17 items

Controllers monitor separate voice channels covering the flight director loop, spacecraft communications, and technical support backrooms. Tapes preserve clipped verbal cadences, overlapping readouts, and background hum from tracking ground stations. Transmission fidelity changes when signals switch between worldwide receiver sites. Transcripts document immediate verbal responses to warning tones and telemetry drops.

ENGINEERING DIVISION DEMOGRAPHICS · 15 items

The flight controllers on duty are largely young civilian engineers in their twenties and thirties. Many staff members hold engineering degrees from regional universities across the American South and Midwest. Several positions are filled by military veterans with systems maintenance backgrounds. The room operates through assigned stations including flight dynamics, guidance, and environmental control.

TECHNICAL CRISIS VOCABULARY · 10 items

Flight teams speak in abbreviated operational acronyms to manage telemetry data quickly. Controllers refer to flight positions as Trench, GUIDO, and CAPCOM during anomalies. Numerical values for power budgets and consumable percentages are called out in standard short-form code. Telemetry failure reports follow rigid verbal checklists established in malfunction procedure manuals.

OFFICE OF CIVILIAN DEFENSE RELATIONS · 12 items

Federal security details coordinate with local police departments to protect the space center perimeter. Armed guards monitor gate traffic and control access points around sensitive technical buildings. Administrative offices maintain contact records with civilian law enforcement during the Vietnam War period. Security officers log perimeter disturbances and visitor movement around federal assets.

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MANUAL CALCULATION TOOLS · 14 items

Engineers rely on manual calculation tools when mainframe telemetry channels drop offline. Controllers calculate burn times and orbital changes using Pickett pocket slide rules. Teams mark reference trends on printed grid paper using grease pencils. Desk calculators and printed trajectory tables provide manual verification of remaining electrical reserves.

REAL-TIME TELEMETRY DATA FLOW · 14 items

Spacecraft tracking data flows from overseas antennas into the Goddard Space Flight Center in Maryland. Mainframe computers process incoming telemetry signals and route them directly to Houston. High-speed lines drive the numeric displays on console screens and front-wall plotters. Technicians track real-time coordinates against fixed mission trajectory curves.

Casting

ENGINEERING DIVISION DEMOGRAPHICS · 15 items

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Costume

SHIFT ROTATION ATTIRE · 27 items

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cycles.



Art direction

CONSOLE HARDWARE ARCHITECTURE · 16 items

Building 30 contains two Mission Operations Control Rooms on the second and third floors. Consoles house cathode-ray tube displays alongside illuminated push-button indicators for systems monitoring. Pneumatic tube stations move paper telemetry summaries and schematics across the room. Overhead projection screens display spacecraft coordinates and trajectory plots above the console rows.

CONTROL ROOM CONSUMABLES · 38 items

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CLEAR LAKE COMMUTING · 12 items

Controllers commute to the Clear Lake complex along local Texas access highways and regional roads. Personnel park personal sedans and pickups in open asphalt lots outside the security perimeter. Security personnel check identification credentials at gated access booths. The facility sits isolated twenty-five miles southeast of downtown Houston.

PRESS CORPS INFRASTRUCTURE · 8 items

Press operations occupy the auditorium within Building 1 adjacent to the control facility. Television cameras sit on elevated wooden risers facing the main presentation stage. Rows of teletype machines and banks of coin-operated payphones line the media workspace. Briefing officers distribute paper mission change sheets to reporters after major shifts.

SUPPORT STAFF TRENCH DYNAMICS · 32 items

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Sound

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Cinematography

CONSOLE HARDWARE ARCHITECTURE · 16 items

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Locations

CLEAR LAKE COMMUTING · 12 items

Controllers commute to the Clear Lake complex along local Texas access highways and regional roads. Personnel park personal sedans and pickups in open asphalt lots outside the security perimeter. Security personnel check identification credentials at gated access booths. The facility sits isolated twenty-five miles southeast of downtown Houston.

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PRIOR ART

65 existing films surveyed for a similar premise, seeded from: space crisis, mission control, engineers, survival, 1970s.

Apollo 13 (1995)

exploited mechanical flaw in the oxygen system · victim · triumphant · professional duty overcomes disaster · crew rescued after safe splashdown

similarity 0.72 · <https://www.themoviedb.org/movie/568>

Salyut-7 (2017)

exploited electrical flaw in station hardware · investigator · heroic · loyalty to comrades redeems political coercion · station restored to operational capability

similarity 0.66 · <https://www.themoviedb.org/movie/438740>

The Right Stuff (1983)

exploited psychological flaw in institutional hubris · bystander · satirical · individual audacity transcends bureaucratic manipulation · pioneering era closes with success

similarity 0.66 · <https://www.themoviedb.org/movie/9549>

The Real Right Stuff (2020)

exploited historical record shipped as archival media · bystander · documentary · collective effort secures a national achievement · historical missions conclude successfully

similarity 0.64 · <https://www.themoviedb.org/movie/753325>

First Man (2018)

exploited emotional flaw of unexpressed grief · victim · somber · private sorrow finds release through professional endurance · mission completed with silent personal reconciliation

similarity 0.64 · <https://www.themoviedb.org/movie/369972>

The Martian (2015)

exploited scientific ingenuity shipped as standard equipment · victim · optimistic · relentless problem-solving affirms human resilience · stranded astronaut successfully retrieved

similarity 0.63 · <https://www.themoviedb.org/movie/286217>

Hidden Figures (2016)

exploited systemic bias flaw in workplace hierarchy · investigator · inspiring · competence forces institutional integration · ground calculations validate the orbital trajectory

similarity 0.62 · <https://www.themoviedb.org/movie/381284>

Love (2011)

exploited psychological isolation flaw in prolonged confinement · victim · melancholic · alienation deepens into surreal acceptance · isolated subject achieves transcendence

similarity 0.58 · <https://www.themoviedb.org/movie/54320>

First Man into Space (1959)

exploited biological mutation flaw caused by cosmic radiation · perpetrator · tragic · reckless ambition brings monstrous ruin · dying monster provides crucial scientific data

similarity 0.56 · <https://www.themoviedb.org/movie/28955>

The Devil's Mouth (2026)

exploited physical vulnerability flaw in a trapped environment · victim · dread · survival pressure fractures ethical solidarity · diminished survivors escape the predator

similarity 0.55 · <https://www.themoviedb.org/movie/1481343>

Santo vs. the Martian Invasion (1967)

exploited alien invasion feature shipped as advanced weaponry · investigator · camp · physical heroism preserves terrestrial security · alien invaders suffer total defeat

similarity 0.55 · <https://www.themoviedb.org/movie/80766>

It All Adds Up - The Making of Hidden Figures (2017)

exploited promotional commentary shipped as retrospective access · bystander · celebratory · historical recovery honors overlooked labor · filmmakers validate historical relevance

similarity 0.55 · <https://www.themoviedb.org/movie/1533402>

Where this stands, and where it does not yet

Originality is claimed only relative to the titles above, never absolutely.

The narrative adopts the perpetrator point of view inside Mission Control, focusing on flight controllers whose previous procedural oversights caused the technical failure.

None of the listed titles present the disaster strictly through the culpability of the ground controllers.

checked against: Apollo 13, Salyut-7, The Right Stuff, The Real Right Stuff, First Man, The Martian, Hidden Figures, Love, First Man into Space, The Devil's Mouth, Santo vs. the Martian Invasion, It All Adds Up - The Making of Hidden Figures

→ Frame the central flight controller as the engineer who signed off on the compromised oxygen tank specifications.

The narrative operates as an institutional tragedy ending in the irreversible destruction of administrative careers despite the survival of the flight crew.

None of the listed films conclude with bureaucratic self-destruction alongside a successful rescue.

checked against: Apollo 13, Salyut-7, The Right Stuff, The Real Right Stuff, First Man, The Martian, Hidden Figures, Love, First Man into Space, The Devil's Mouth, Santo vs. the Martian Invasion, It All Adds Up - The Making of Hidden Figures

→ Show ground command securing splashdown while an internal audit terminates their program authority.

The narrative adopts a tone of relentless bureaucratic dread, focusing exclusively on ground support personnel paralyzed by statistical inevitability.

None of the listed historical or space flight films use an sustained tone of pure procedural dread without heroic relief.

checked against: Apollo 13, Salyut-7, The Right Stuff, The Real Right Stuff, First Man, The Martian, Hidden Figures, Love, First Man into Space, The Devil's Mouth, Santo vs. the Martian Invasion, It All Adds Up - The Making of Hidden Figures

→ Confine the perspective entirely to telemetry analysts convinced their models predict crew asphyxiation.

RESEARCH LOG

What each pass searched, found, and did next. Process detail, not required reading.

PASS 000

114 sources · 133 fragments · 40 via Extract

coverage 0% → 82% confidence 0% → 77%

CONSOLE HARDWARE ARCHITECTURE — Apollo Mission Operations Control Room MOCR console schematics Philco Ford CRT pneumatic tube; NASA MSC Building 30 MOCR console hardware display controls layout technical report

COMMUNICATIONS LOOP ACOUSTICS — Apollo 13 flight director loop capcom audio recordings transcript April 1970; NASA Apollo 13 internal communications voice loop trench backroom audio tapes archive

ENGINEERING DIVISION DEMOGRAPHICS — Apollo 13 flight controllers roster biographies educational background military service; NASA MSC Flight Operations Division personnel demographics flight controllers 1970

SHIFT ROTATION ATTIRE — NASA Mission Control Apollo 13 high resolution archival photographs controllers MOCR April 1970; Apollo 13 mission control room shift rotation candid photos controllers clothing rumpled

CONTROL ROOM CONSUMABLES — NASA Mission Control center Building 30 cafeteria vending coffee brands tobacco 1960s 1970; Apollo 13 mission operations control room ashtrays coffee cups takeout food archival photos

TECHNICAL CRISIS VOCABULARY — Apollo 13 telemetry acronyms abbreviations flight controller jargon mission operational terms; NASA Apollo flight controller terminology life support EPS EECOM telemetry shorthand handbook

FACILITY LIGHTING PROTOCOLS — MSC Building 30 Mission Operations Control Room lighting levels foot-candles display illumination specifications; Apollo MOCR rear projection screens ambient illumination lighting protocols technical manual

CLEAR LAKE COMMUTING — NASA Manned Spacecraft Center Clear Lake employee parking lot gate security vehicles 1970; Clear Lake Houston NASA Road 1 commute personnel parking cars April 1970 historical photos

PRESS CORPS INFRASTRUCTURE — MSC Building 1 auditorium news center Apollo 13 press corps floor plan teletype; Apollo 13 press room television camera positions NASA Manned Spacecraft Center newsroom setup 1970

OFFICE OF CIVILIAN DEFENSE RELATIONS — NASA Manned Spacecraft Center security procedures Vietnam War antiwar protests 1970; MSC Houston perimeter security police civil defense coordination NASA complex April 1970 records

→ investigate CONTROL ROOM CONSUMABLES (score 0.31)

PASS 001

36 sources · 34 fragments · 12 via Extract

coverage 82% → 82% confidence 77% → 76%

opened: SUPPORT STAFF TRENCH DYNAMICS

SHIFT ROTATION ATTIRE — Apollo 13 Mission Operations Control Room high resolution archival photos controllers April 1970; NASA flight controllers clothing shirts ties fatigue Apollo 13 photographs shift changes

CONTROL ROOM CONSUMABLES — Apollo 13 Mission Control coffee cups cigarette brands food containers April 1970; NASA MOCR consoles food drink ashtrays vending snacks Apollo missions archive photos

SUPPORT STAFF TRENCH DYNAMICS — NASA Apollo Mission Control Staff Support Room backroom layout procedures trench; Apollo 13 RETRO FIDO backroom trajectory calculation procedures support staff organizational chart

→ investigate SUPPORT STAFF TRENCH DYNAMICS (score 0.26)

PASS 002

38 sources · 32 fragments · 12 via Extract

coverage 82% → 83% confidence 76% → 75%

opened: MANUAL CALCULATION TOOLS, SUPPORT STAFF TRENCH DYNAMICS

CONTROL ROOM CONSUMABLES — Apollo 13 Mission Operations Control Room MOCR coffee cups cigarettes food trash photographs April 1970; NASA MSC Building 30 Mission Control consumables vending machines ashtrays brands 1970

SUPPORT STAFF TRENCH DYNAMICS — NASA Apollo Staff Support Room backroom layout procedures Trench FIDO GUIDO flight dynamics; Apollo 13 Mission Operations Control Room Staff Support Room organizational hierarchy communication loop

MANUAL CALCULATION TOOLS — Apollo 13 mission control slide rules calculators grease pencils trajectory manual calculation tools; NASA MOCR Apollo flight controller slide rule model Friden calculator graphing paper inventory 1970

→ investigate SUPPORT STAFF TRENCH DYNAMICS (score 0.26)

PASS 003

41 sources · 33 fragments · 12 via Extract

coverage 83% → 85% confidence 75% → 76%

opened: REAL-TIME TELEMETRY DATA FLOW, SUPPORT STAFF TRENCH DYNAMICS

CONTROL ROOM CONSUMABLES — Apollo 13 Mission Operations Control Room food coffee cigarettes photos; NASA Mission Control Center building 30 consumables vending coffee cups 1970

SUPPORT STAFF TRENCH DYNAMICS — Apollo 13 Staff Support Room backroom flight dynamics trench organizational hierarchy; Apollo mission control SPAN Trench backroom procedures calculations documentation

REAL-TIME TELEMETRY DATA FLOW — Apollo telemetry data flow MSFN Goddard Real-Time Computer Complex MOCR displays; NASA Manned Spacecraft Center Real Time Computer Complex RTCC CRT display generation 1970

→ investigate SUPPORT STAFF TRENCH DYNAMICS (score 0.26)